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SMART AGRICULTURE SYSTEM

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Group members

London Met ID	Student Name
24058830	Adhish Bhandari
24058802	Abhyudaya Narayan Shrestha
24058837	Raunak Thapa Magar
24058824	Upashya Pathak

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Acknowledgement

We are deeply grateful to our teacher, Mr. Jaganath Paudyal, and our module leader, Mr. Shishir Subedi sir, for providing us with this invaluable opportunity to work on an IoT project and for their unwavering support while we wrote a report. We also want to express our gratitude to every member of our team for their work and collaboration, which made this report possible. Our achievement was largely due to the advice of our mentors.

Abstract

The creation of an intelligent irrigation system to assist small-scale farmers is the main objective of this project. Using real-time data, the goal is to automate irrigation chores when the farmer is not there. A relay module for controlling a water pump, temperature sensors, soil moisture sensors, and ESP32 are some of the parts of the system. To enable online monitoring of sensor readings, the software side relates to the Blynk IoT platform. The pump is activated by the system to water the crops when the soil moisture falls below a predetermined threshold. Both a PC and a smartphone can examine the data collected.

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1. Introduction

1.1. Explanation of IoT Technology

The internet of Things, also known as IoT, is the network of physical devices that connect with each other where data is collected, shared and processed in real time and act upon it. Household items like Thermostats and Smoke Detectors are some examples of this technology. IoT technology works by collecting data from sensors to gather data like Temperature, Humidity, or Movement. Then the data is transmitted using Wi-Fi or Bluetooth to Cloud Systems or Local Platforms. The collected data is then analyzed and processed which can be monitored and given instructions to execute on Apps or Websites. IoT in agriculture is composed of sensors to check soil moisture for detection of the condition of soil and other components to water the crops and detect unwanted obstruction in the farm/soil (Mushtaq, 2023) (Spotta, 2025).

1.2. Current Scenario of Agriculture: Global and Nepal Context

Agriculture is performed by twenty seven percent of the global population (Nijkamp, 2000). It is one of the major occupations of the world as it plays a huge role in feeding eight billion people. However, it is facing new issues like Abrupt Climate Change, High Resource Scarcity, and Rapid Population Growth. Devices like Soil Sensors, Watering Drones, and Temperature Sensors optimize the production of various crops and resolve the high use of pesticides (Mushtaq, 2023). The Global Market of IoT in Agriculture is huge and has not reached its full potential, especially in developing countries due to lack of budget and infrastructure (Grand View Research, 2024).

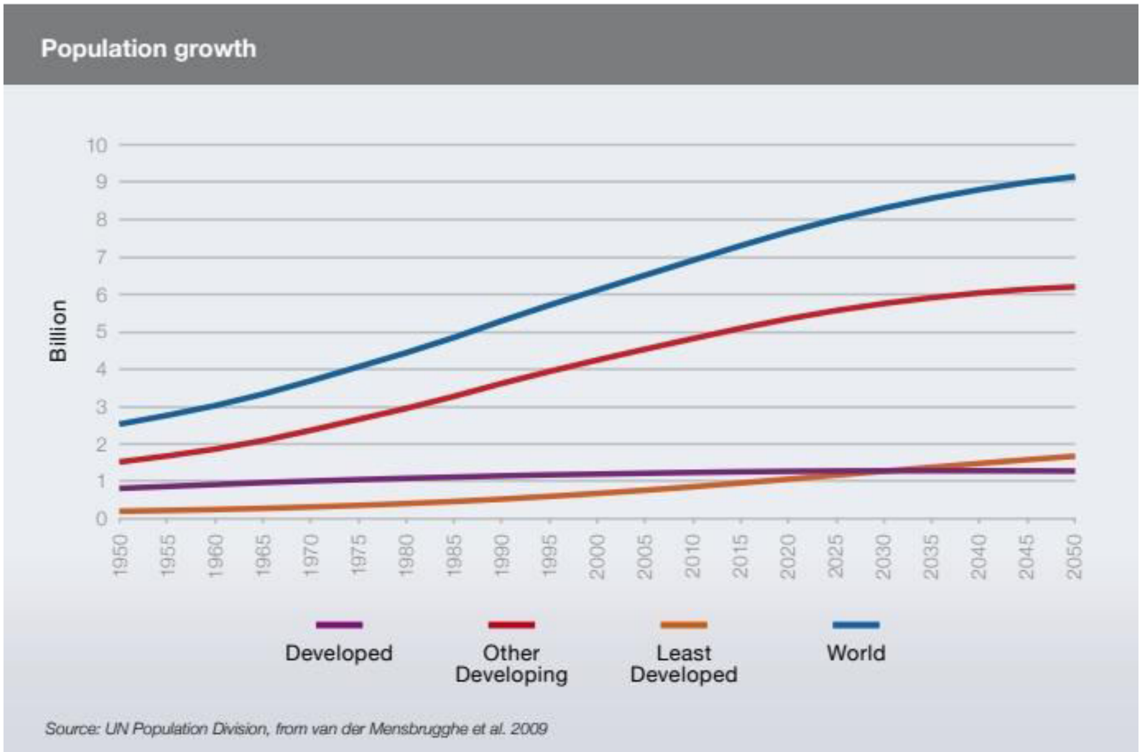

Figure 1: Population Growth

Nepal's economy runs on agriculture as sixty two percent of Nepalese employ themselves in agriculture and contribute twenty three percent to nations GDP. IoT Technologies have the potential to modernize and magnify Nepali agriculture by enabling real time monitoring of soil, water, and crop health. However, adopting this technology is difficult due to High Costs, as devices are expensive for smaller scale farmers, and Digital Illiteracy is low as many farmers lack skill to advanced tools (Lamsal, 2023). Currently, only a small portion of Nepali farmers use IoT, but initiatives like Digital Ag Nepal are working to promote technology by sharing innovations and addressing digital knowledge and training (GeoKrishi, 2024). Various studies suggest that tailored IoT solutions, supported by expert consultation and data collection result into significant improvement in resource management and yield in Nepal's diverse agricultural spectrum (Yadav, 2023).

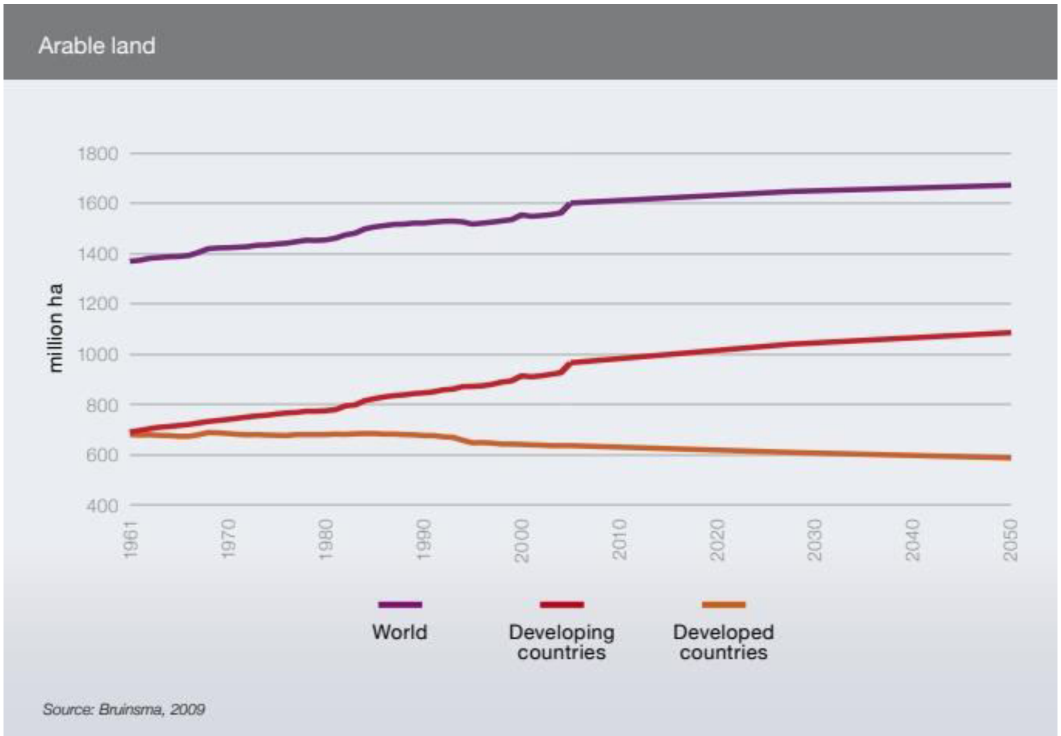

Figure 2: Airable Land

1.3. Problem Statement and Project Solution

Forced use of traditional machinery because of lack of development, modern agriculture technology is a huge cause of low productivity seen in the farming industry. With rising sea levels and overall increase in global temperatures, sudden weather changes have disrupted harvesting patterns, ultimately damaging crop production (Giri, 2023). To lower such issues, implementation of an IoT-based monitoring system can be done to address these problems in real time and provide data about the same accordingly. Farmers can remotely monitor their farms in real time and control them using IoT systems to address any issues and make quick decisions accordingly. Common issues such as low moisture level in soil or excessive presence of insecticides and pesticides can be monitored and dealt with consequently (Mushtaq, 2023). All in all, the problems that are seen in agriculture today can be addressed using the IOT integrated system to facilitate different farming activities.

1.4. Aims and Objectives

The aim of this coursework is to implement modern agricultural farming practices using an IoT system. The system should be autonomous in its approach and do most of the things automatically that would have otherwise been done manually by a farmer. The overall system should be sustainable, scalable, and adaptable with the prime goal of increasing efficiency. Implement automated irrigation systems that check soil moisture levels and water plants accordingly. Adapt watering schedules based on temperature and humidity levels to address unexpected weather conditions. Checking water supplies on storage tanks to prevent shortages and optimize water usage. Allowing farmers to remotely monitor and control their farms through a user interface and send notifications and alerts in case of external damage to crops (Yadav, 2023) (Mushtaq, 2023).

2. Background

2.1. System Overview

The smart farm system consists of various sensors and actuators to work in collaboration with ESP-32 as the microcontroller platform to orchestrate integration. The system will overview the climatic conditions to make data driven decisions for the crops while soil monitoring will be done to ensure the best health of the soil. Visual annotations and alerts will also be given both through cloud and local computing methods. The system will also have a client end where the user can get visualization of the data, time series and statistics alongside a local LCD to display Real Time data from the sensors. The architecture will also automate a lot of redundant tasks such as watering the crops and opening fences for farmers, and many more. These tasks will be performed by suitable actuators and sensors being controlled by the user or automatically programmed on the microcontroller. The ESP32 will use its inbuilt Wi-Fi module to communicate with Blynk for the user end processes through MVC architecture.

2.2. Design Diagram

2.2.1. System Architecture

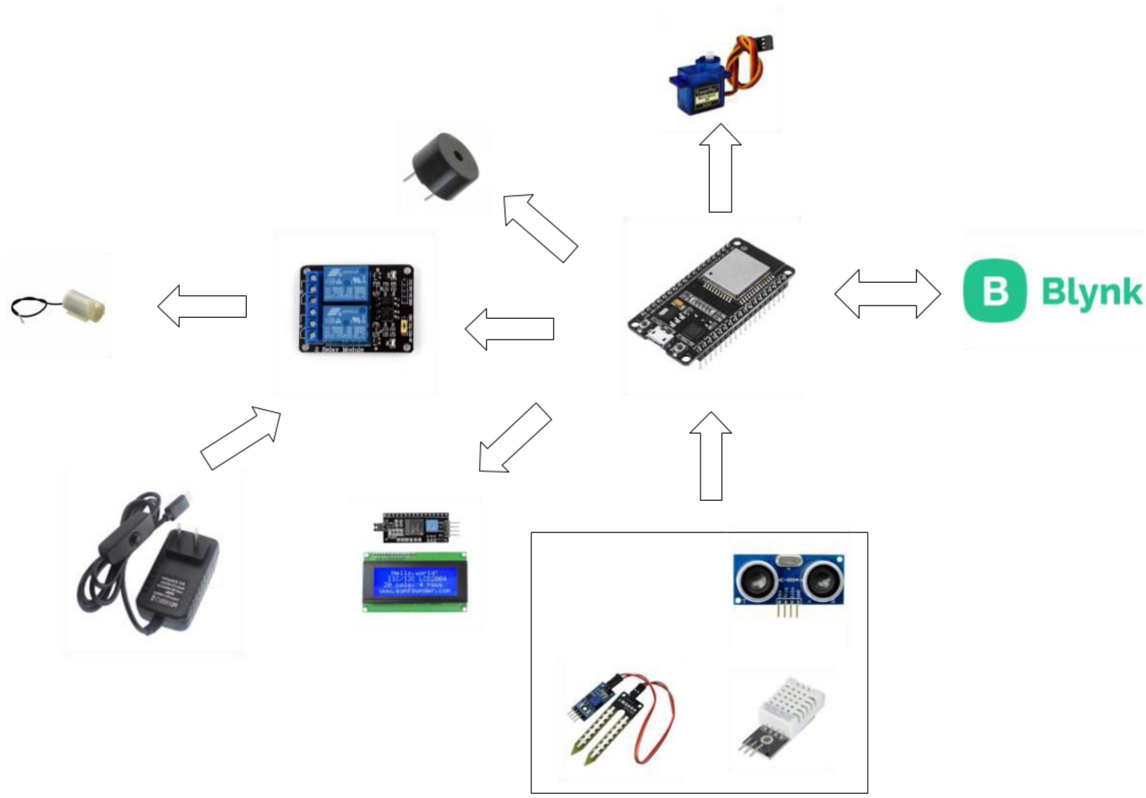

Figure 3: System Architecture

2.2.2. Flowchart

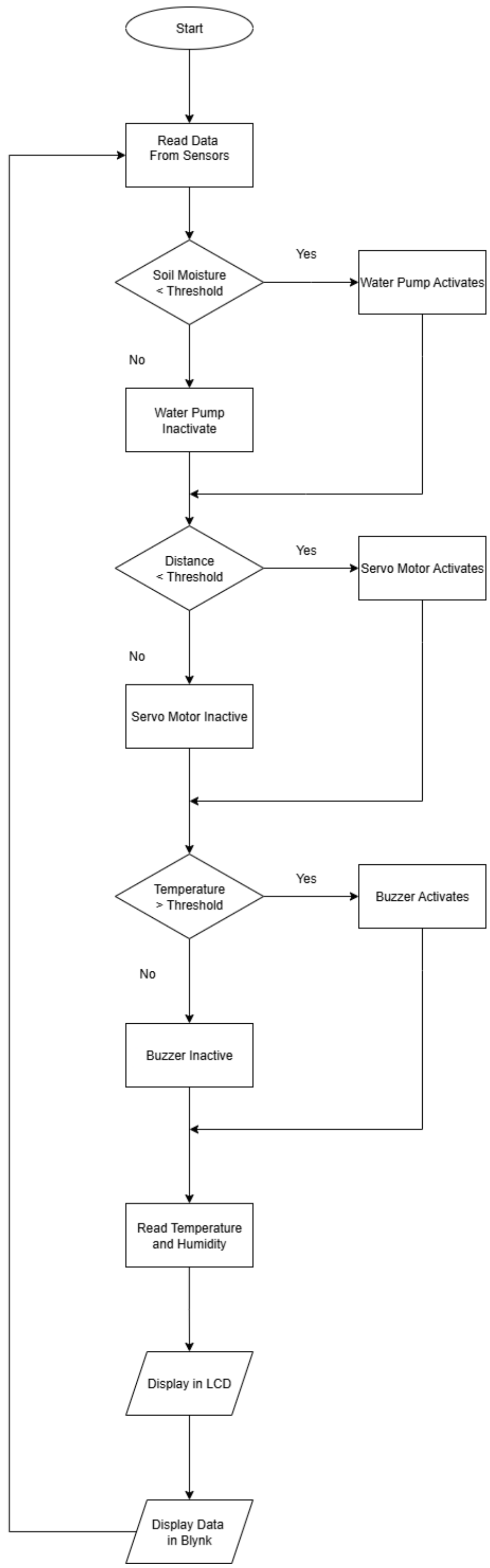

Figure 4: Flowchart

2.2.3. Circuit Diagram

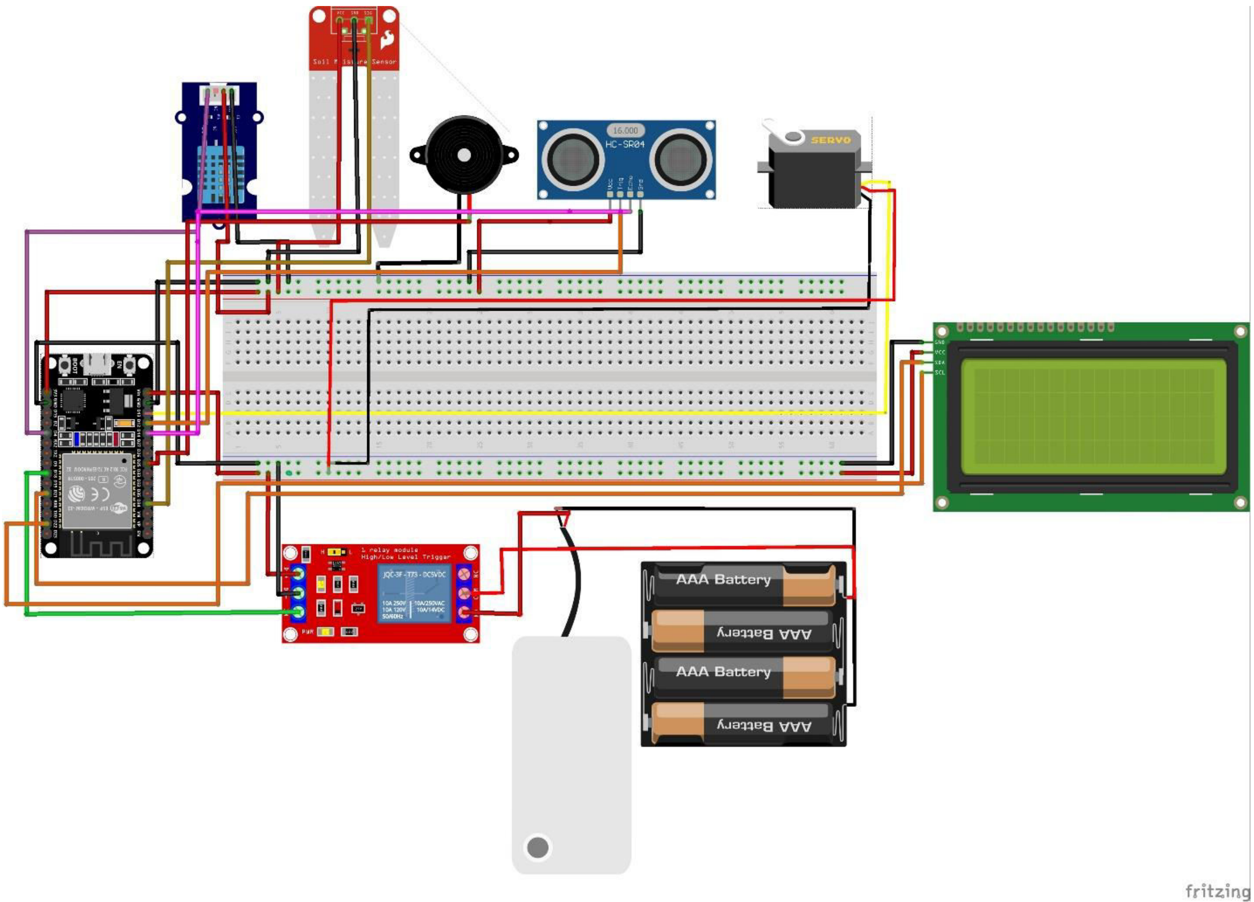

Figure 5: Circuit Diagram

2.3. Requirement Analysis

The system will require DHT22 (Temperature & Humidity) sensor for getting the temperature and humidity data collection, Soil Moisture Sensor for monitoring the soil's moisture level, Ultrasonic Sensor (HC-SR04) to measure the water level in the tank, LCD Display IC2 for displaying the locally computed data like soil moisture and temperature along with humidity, Relay to give the required power to the water pump by amplifying the 5v from the ESP32 rail from bread board, Buzzer for alerting in any case of anomaly, ESP32 as the central node to control and process the sensors and actuators along with connection for Blynk cloud system, Breadboard for clean connection arrangement and power supply management and 5V 2A adapter to power the ESP32, sensors and actuators. These were the hardware requirements, now for the software requirement we will be needing Arduino IDE to compile and boot the ESP32 with the required code to make the IoT system seamless integrated, Blynk for cloud processes and user end controls, Fritzing for circuit design and schematics.

2.3.1. Hardware Requirements

2.3.1.1. ESP-32

It’s an IoT device that has an on-chip 32-bit microcontroller which is integrated with Wi-Fi and Bluetooth which performs various tasks.

Figure 6: ESP-32

2.3.1.2 LCD

It's an IoT device that displays output especially used in calculators, phones and computers.

Figure 7: LCD

2.3.1.3 Ultrasonic Sensor

It's an IoT device which is suitable to detect objects or liquids amid other presences.

Figure 8: Ultrasonic Sensor

2.3.1.4. Soil Moisture Sensor

It's an IoT device which measures the moisture or the water level present in the soil.

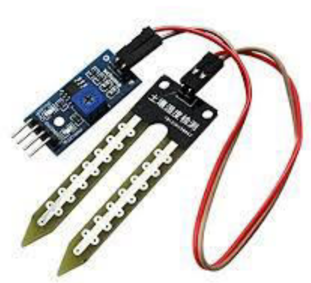

Figure 9: Soil Moisture Sensor

2.3.1.5. Double Relay

It's an IoT device that allows small amounts of current to control higher current.

Figure 10: Double Relay

2.3.1.6. Buzzer

It’s an IoT device that produces buzzing sound when activated using current.

Figure 11: Buzzer

2.3.1.7. Breadboard

It’s an IoT device that is used as a base for constructing electronic circuits.

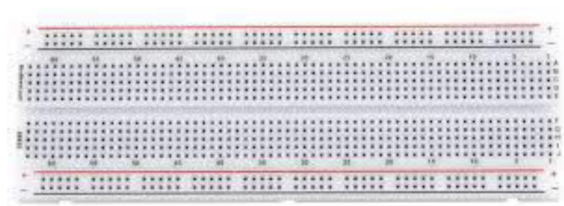

Figure 12: Breadboard

2.3.1.8. Water Pump

It's an IoT device that is required to disperse water in certain projects.

Figure 13: Water Pump

2.3.1.9. 5V Adapter

It's an IoT device that converts AC to 5V DC used to power low voltage devices.

Figure 14: 5V Adapter

2.3.1.10. Jumper Wires

It's an IoT device that is a wire which consists of connecters and conduciveness to develop electrical connection.

Figure 15: Jumper Wires

2.3.1.11. DHT Sensor

It's an IoT device that measures both temperature and relative humidity in the surrounding air.

Figure 16: DHT Sensor

2.3.1.12. Servo Motor

Its an IoT device that converts the signal of a microcontroller into rotational angular displacement or angular velocity of the motion output shaft.

Figure 17: Servo Motor

2.3.2. Software Requirements

2.3.2.1 Arduino IDE

It’s an IoT program that is to write codes and execute for IoT devices.

Figure 18: Arduino IDE

2.3.2.2 Blynk

It's an IoT program that easily manages and monitors IoT devices from anywhere.

Figure 19: Blynk

2.3.2.3 Fritzing

It's an IoT program that helps to design the prototype of electronic circuits.

Figure 20: Fritzing

3. Case Study

3.1. Introduction

In this project, we aim to show the possible use of IoT in agriculture. Agriculture is facing challenges such as water scarcity, low crop yields, and not enough production due to traditional methods of farming (Nijkamp, 2000) (Mushtaq, 2023). Nepal, being a country that's highly dependent on agriculture, cannot produce a sustainable number of crops. Our small-scale model of farm will show how a farm can run efficiently and effectively using basic IoT. This approach aligns with the current scenario of Nepal's agricultural state.

3.2. Problem Statement for Case Study

The case study analyses and talks about the problems faced on the inability to develop sustainable agricultural options in Nepal. The country, being heavily dependent upon its natural resources such as food and wood, has contributed to the depletion of soil. Nepal leading to exploitation of natural resources has ultimately decreased soil fertility and led to lower crop yields (Nijkamp, 2000). Additionally, the case study also mentions degrading food quality due to improper systems implemented for irrigation systems and unmanaged use of fertilizers. All these problems lead to excess waste of irrigation sources, crops damaging because of low soil fertility ultimately leading to improper farming. To mitigate such problems, the implementation of a smart IoT based monitoring system will be very beneficial (Yadav, 2023).

3.3. System Analysis

The Smart Agriculture Monitoring System's technical framework remains robust for Nepal, with minor adaptations. Using Sensors including Soil Moisture Sensor, LDR Sensor, and Temperature and Humidity Sensor for collecting real time environmental data. Usage of Microcontroller like ESP32 is used as it has built-in Wi-Fi and Bluetooth module for easy connection to the network. The data collected is sent to cloud services like AWS EC2, which will then process the data and send the required instructions back to the ESP. Ultimately programmed using Arduino IDE (Mushtaq, 2023).

3.4. Suggestions

Solar power could be helpful in locations without consistent electricity. A Nepali software could make it easier for farmers to comprehend the data. If weather updates are included, the system may be able to adjust before rain or dry spells occur. Local agricultural organizations could assist farmers with system setup and provide training. Those who are not tech-savvy could benefit from simple, illustrated guides (Yadav, 2023).

3.5. Conclusion

The farmer will be able to cultivate stronger crops and conserve water thanks to the project (Mushtaq, 2023). Better planning allows them to spend less time on manual labor. If more farmers had access to something like this, it could have helped them. It demonstrates how basic tools can enhance conventional techniques.

4. Development

4.1. Design and Planning

In the beginning the project was virtually visualized in a Fritzing software. The main hardware component used in this project is ESP32, servo motor, LCD, I2C connector, relay, ultrasonic sensor, buzzer, soil moisture sensor and DHT22. It is connected to Blynk software which will help us monitor and control the system.

The components that are connected are given power through the 5v pin and ground of the ESP32 is connected to the bread boards positive and negative terminal. LCD and Relay are connected to the 5v rail through the bread board. The DHT22, Soil moisture sensor, Ultrasonic Distance sensor is connected to the 3.3v rail of the bread board. DHT22 is connected to the G4 pin, soil moisture sensor is connected to the G34 pin. The TRIGGER and ECHO of the ultrasonic sensor is connected to the G12 and G14 pins respectively. The servo motors VCC is connected to the 5v rail of the breadboard, the GND is connected to the ground and PWM is connected to the G13 pin of the ESP32.

4.2. Resource Collection

A letter format was given to us though MST. It was used to request all the components used in our project which are: ESP32, Ultrasonic sensor, DHT22, LCD, I2C, relay, water pump, buzzer, soil moisture.

4.3. System Development

4.3.1. Phase 1

The ESP32 is connected to a bread board. The ground and 5V of the ESP32 are connected to the negative and positive terminal of the bread board respectively.

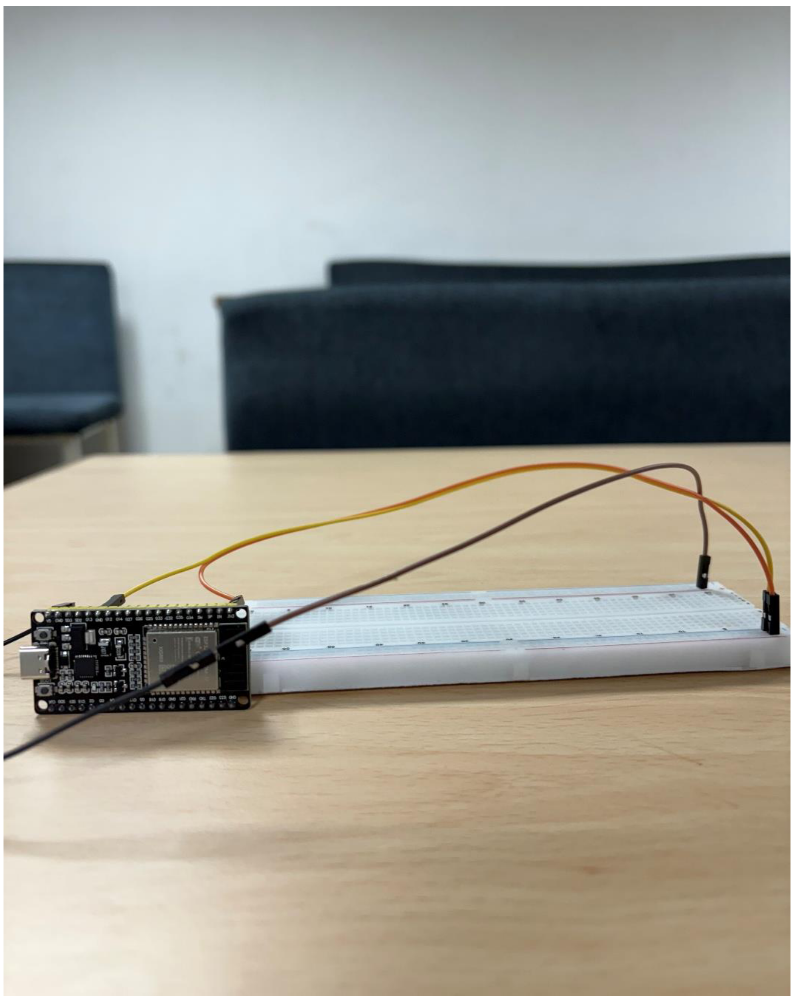

Figure 21: Phase 1

4.3.2. Phase 2

Soil moisture sensor is connected. The VCC pin of the sensor is connected to the 3.3V on the ESP32 through the breadboard. The ground is connected to the negative terminal of the breadboard.

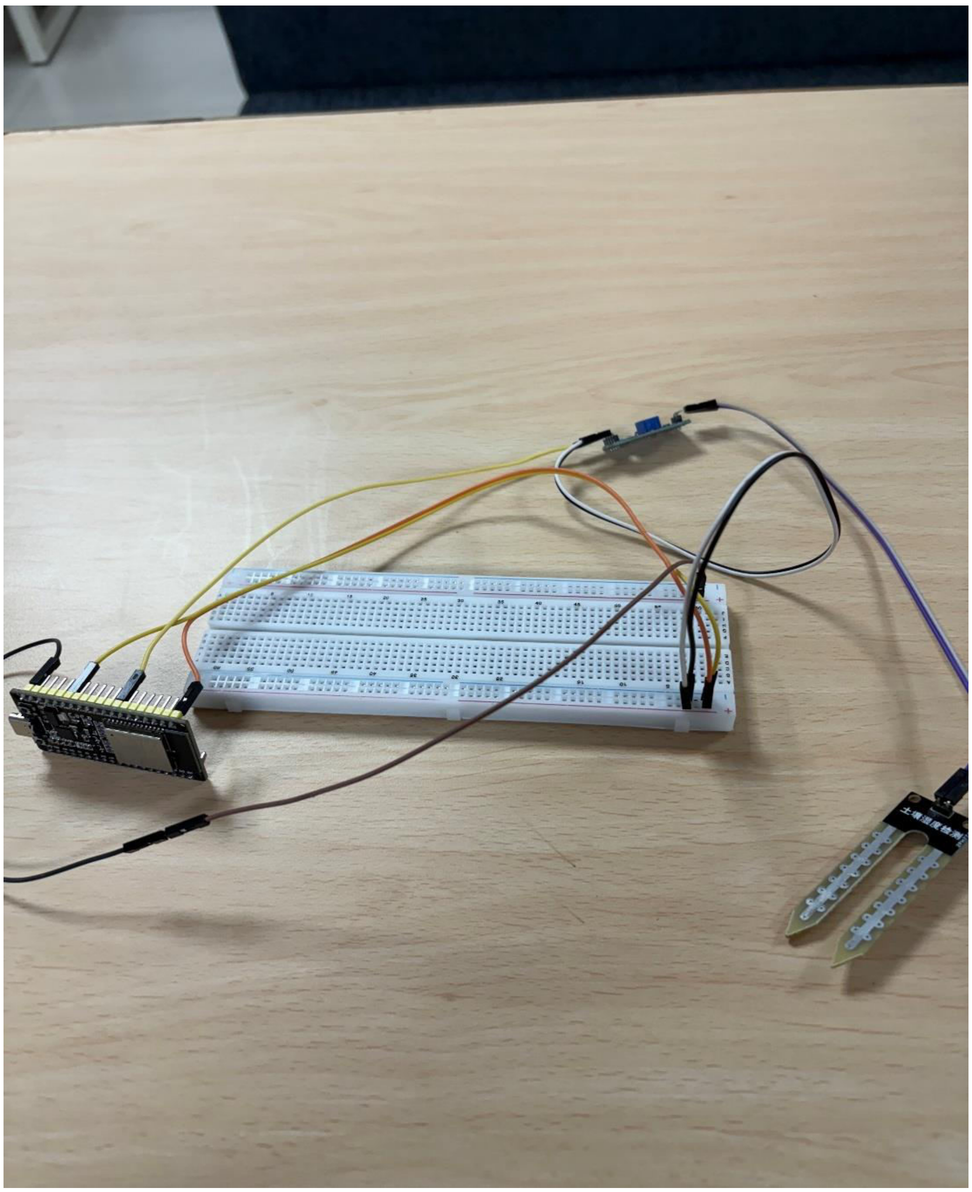

Figure 22: Phase 2

4.3.3. Phase 3

Buzzer is added to the setup. The VCC of the buzzer is connected to GPIO14 on the ESP32. The GND is connected to the ground rail of the breadboard.

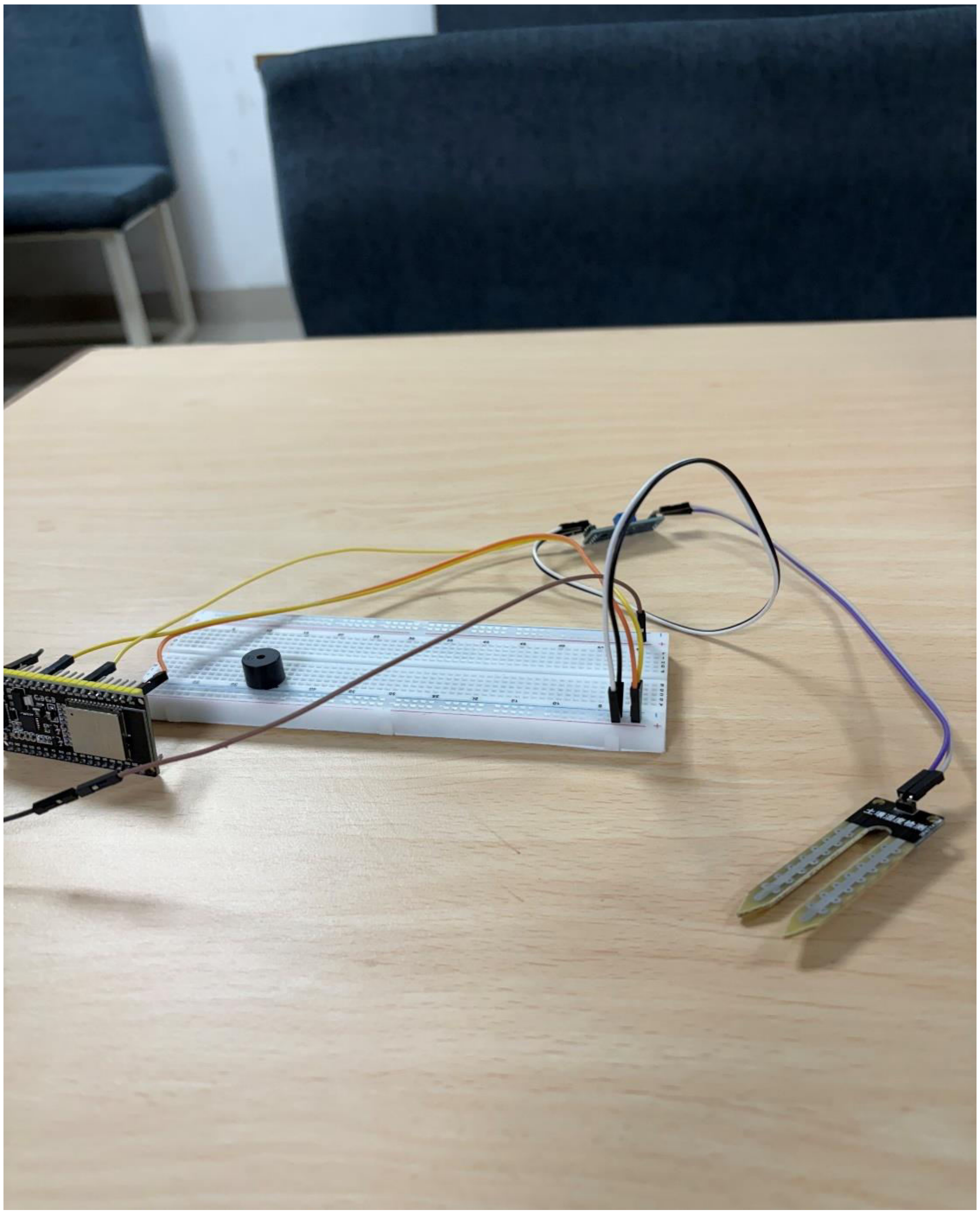

Figure 23: Phase 3

4.3.4. Phase 4

DHT22 sensor is added to the system. The VCC pin of the DHT22 is connected to the 3.3V power rail of the breadboard. The GND pin is connected to the ground rail. The data pin is connected to GPIO4 on the ESP32.

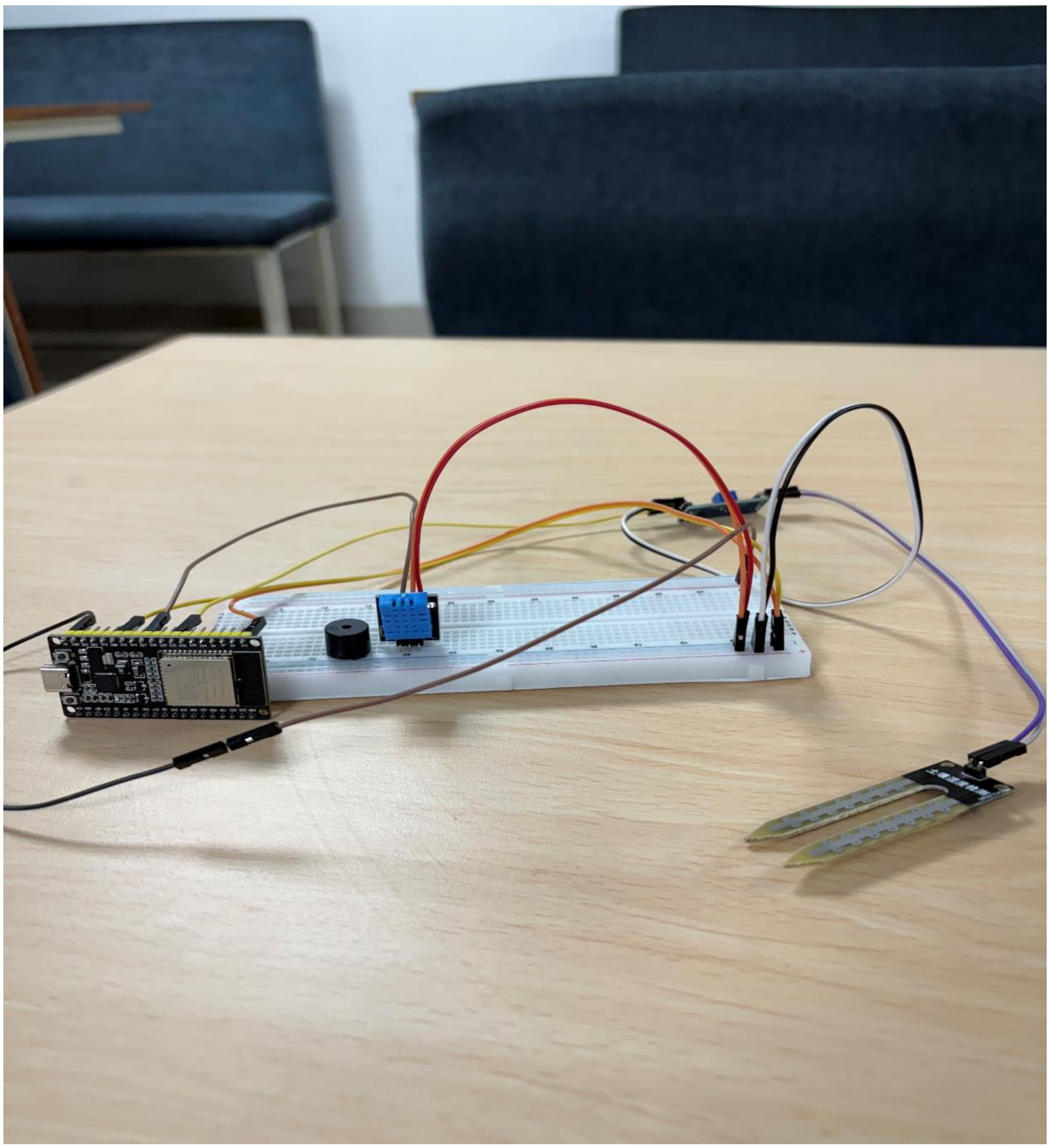

Figure 24: Phase 4

4.3.5. Phase 5

I2C LCD display is integrated into the setup in this phase. The VCC pin of the LCD is connected to the 5V rail on the ESP32 through the breadboard. The GND pin is connected to the ground rail. The SDA line is connected to GPIO21, and the SCL line is connected to GPIO22 on the ESP32.

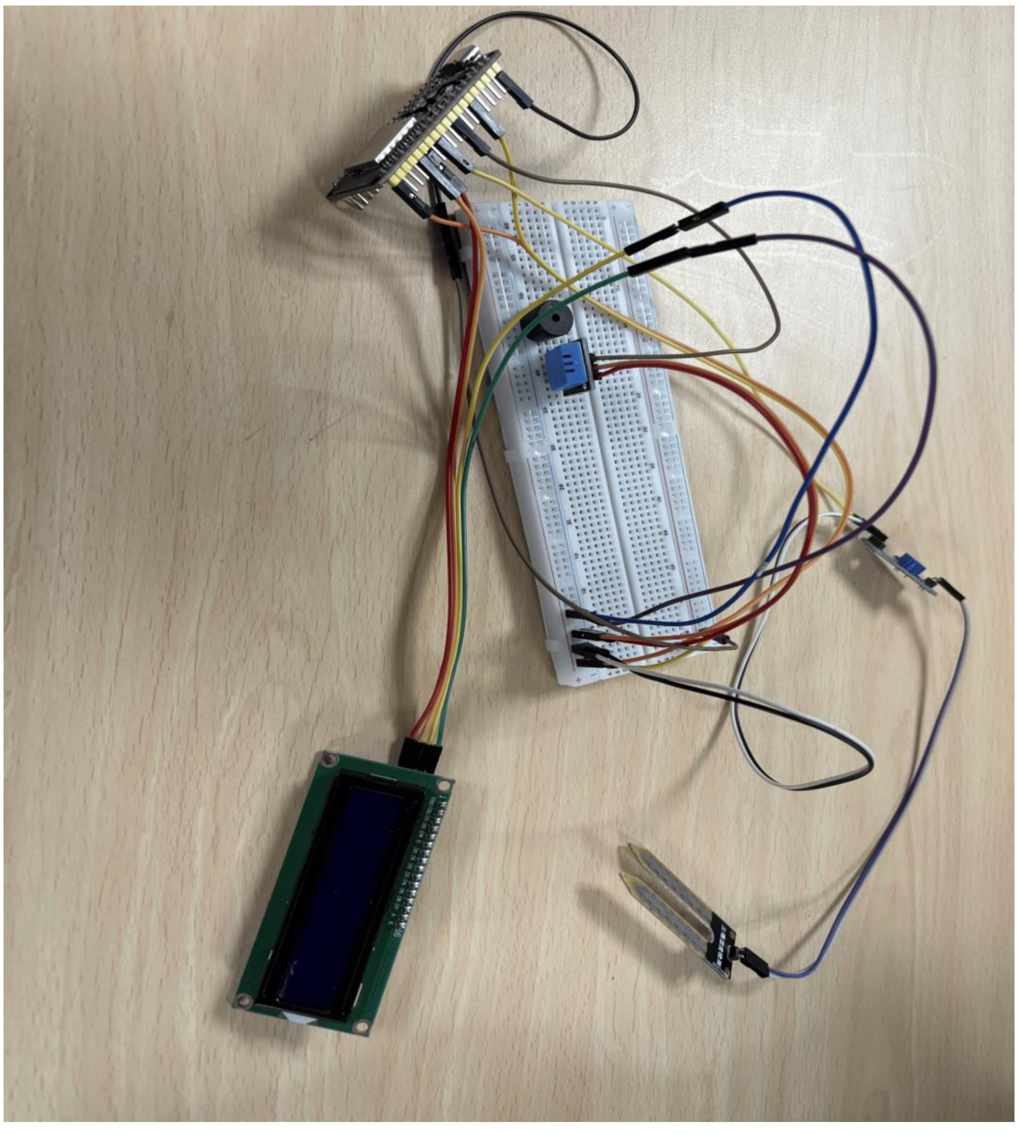

Figure 25: Phase 5

4.3.6. Phase 6

Ultrasonic sensor is added to the circuit in this phase. The VCC pin of the sensor is connected to the 3.3V power rail of the breadboard, while the GND pin is connected to the ground rail. The TRIGGER pin is connected to GPIO26, and the ECHO pin is connected to GPIO25 on the ESP32.

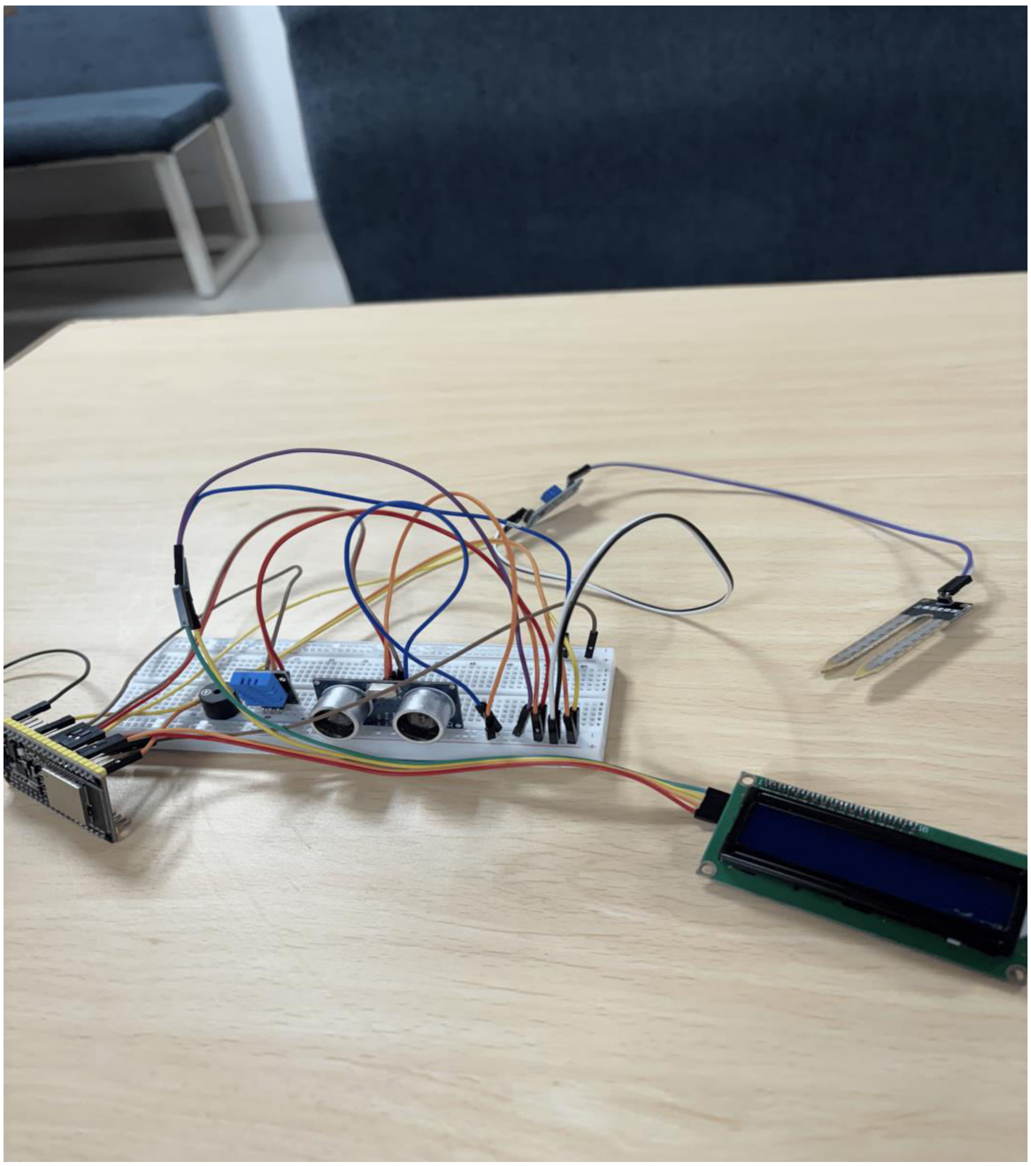

Figure 26: Phase 6

4.3.7. Phase 7

Relay module and a water pump are added to the system, along with a 5V power adapter. The relay module is connected such that the IN1 control pin goes to GPIO27 on the ESP32. The VCC and GND relay are connected to the 5V and GND rails of the breadboard respectively. Water pump's positive terminal is connected to Relay's NO terminal and GND is connected to GND of 5V adapter. 5V adapter's positive wire is connected to COM of the relay.

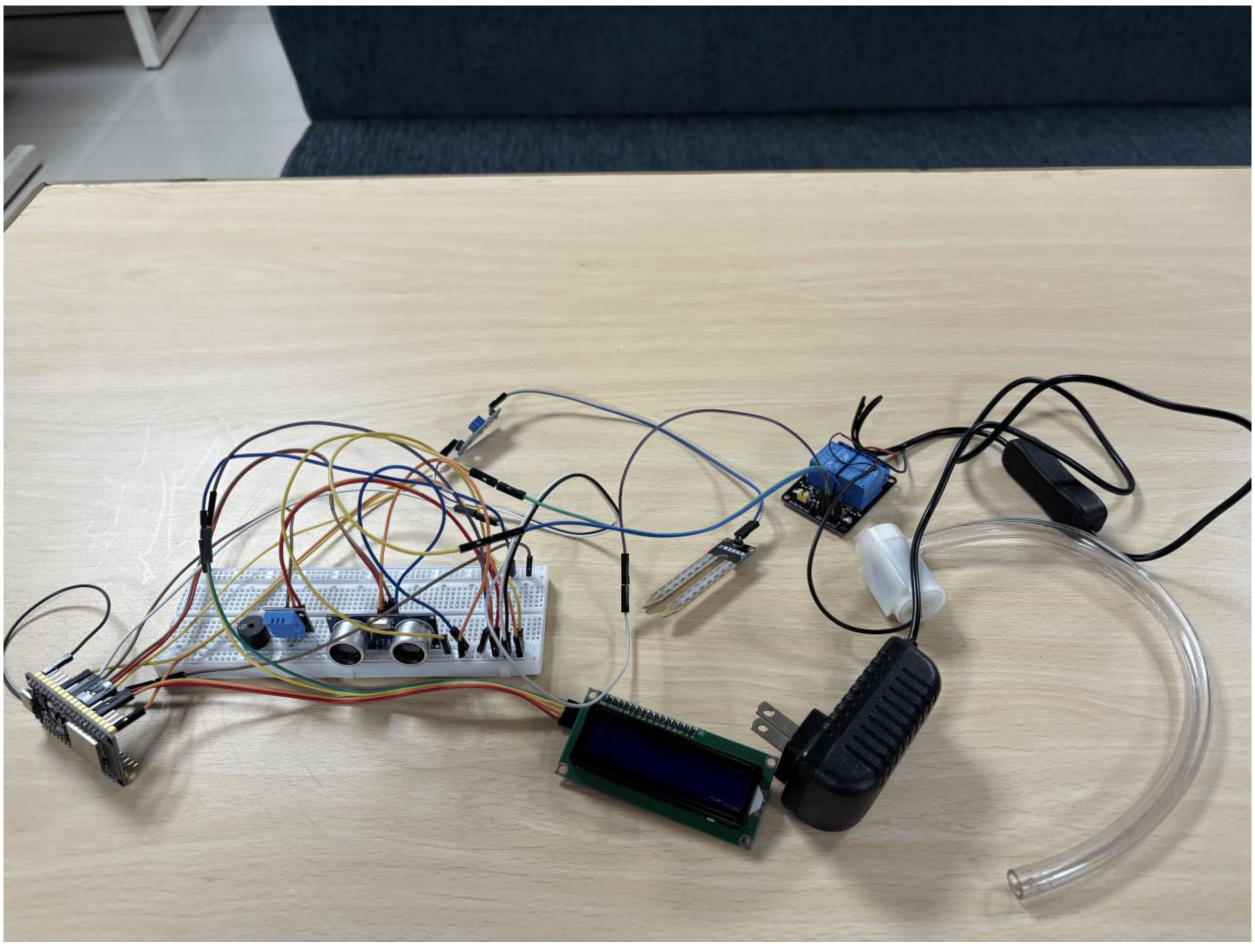

Figure 27: Phase 7

4.4. Pins Connections to Each Components

Component	Pins	ESP32	Bread Board
DHT22	Analogue VCC GND	GPIO 4	VCC (3.3V) GND
Soil Moisture Sensor	Analogue GND VCC	GPIO34	GND VCC (3.3V)
Relay IN1(Pump)	IN1 VCC GND	GPIO27	VCC (5V) GND
Water pump	VCC GND	Controlled via relay VCC->Normally open GND-> GND 5V adapter	
Buzzer	VCC GND	GPIO 14	GND
Ultrasonic Distance sensor	TRIGGER ECHO VCC GND	GPIO 26 GPIO 25	VCC (3.3V) GND
I2C LCD	SDA SCL VCC GND	GPIO 21 GPIO 22	VCC (5V) GND
Servo Motor	VCC GND PWM	GPIO13	VCC(5V) GND

Table 1: Pin Connections

5. Results and Findings

5.1. Test 1: To Check If the System Works Properly or Not.

Test	**1**
Objective	To check if the soil moisture sensor works properly or not.
Activity	The soil moisture analog pin was connected to ESP's Analogue GPIO pin and VCC and GND were connected respectively. The results of the sensor's output were logged into the serial monitor to get the sensor state.
Expected result	When the sensor was in the dry soil the terminal logs the analog read value and soil state as dry. When the sensor was in the wet soil the terminal logs the analogue read value and soil state as moist.
Actual result	When the sensor was in the dry soil the terminal logged the analog read value and soil state as dry. When the sensor was in the wet soil the terminal logged the analogue read value and soil state as moist.
Conclusion	The test was successful

Table 2: Test 1

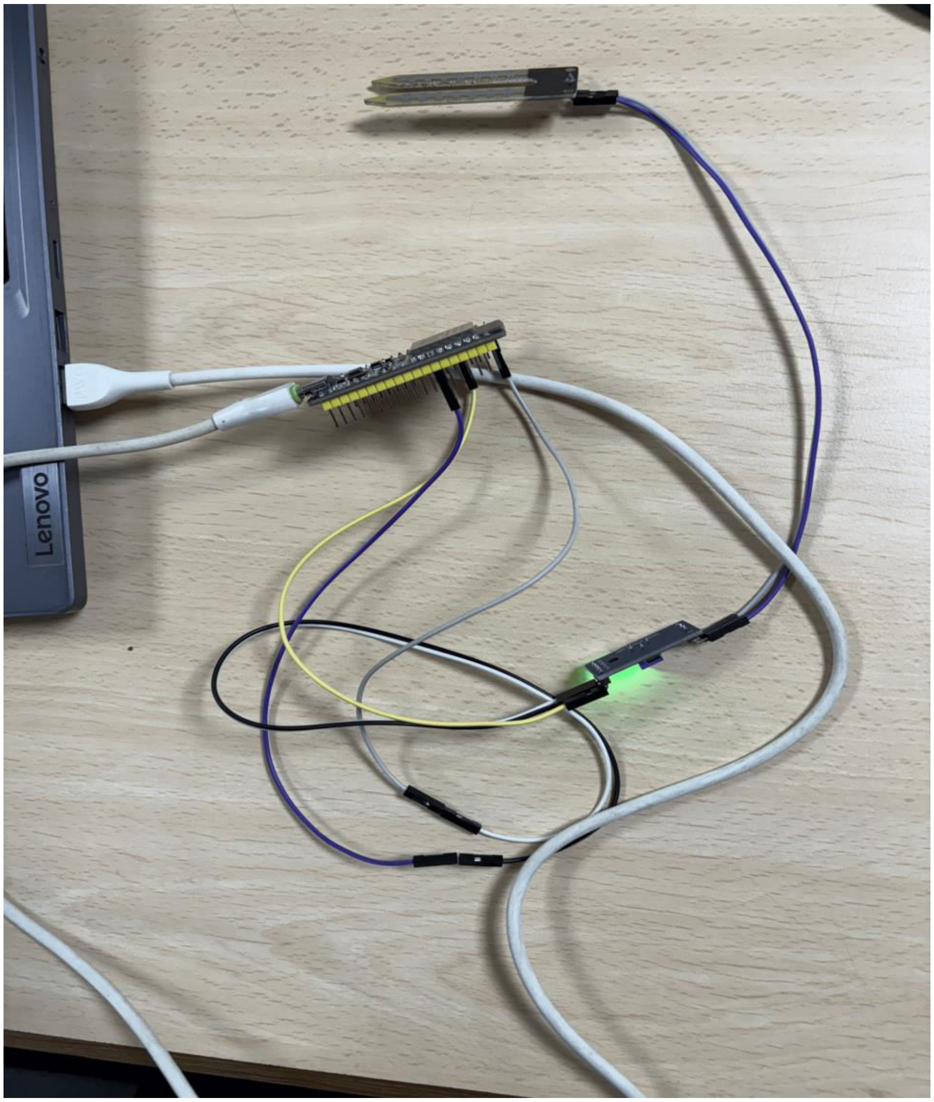

Figure 28: Test 1.1

Figure 29: Test 1.2

5.2. Test 2: To Check If the DHT Sensor Works or Not with LCD.

Test	**2**
Objective	To check if the DHT sensor works properly or not.
Activity	DHT's analog pin was connected to the GPIO pin and the VCC and GND were connected to ESP respectively. The LCD analog pin was also connected to the GPIO pin, the VCC and GND was connected to ESP respectively. The buzzer was also connected to the ESP accordingly.
Expected result	The temperature and humidity readings should be displayed on the LCD, and the buzzer should buzz when temperature is greater than 35$°c$.
Actual result	The readings of temperature and humidity from the DHT were displayed on the LCD, the buzzer buzzed when temperature is greater than 35$°c$
Conclusion	The test was successful

Table 3: Test 2

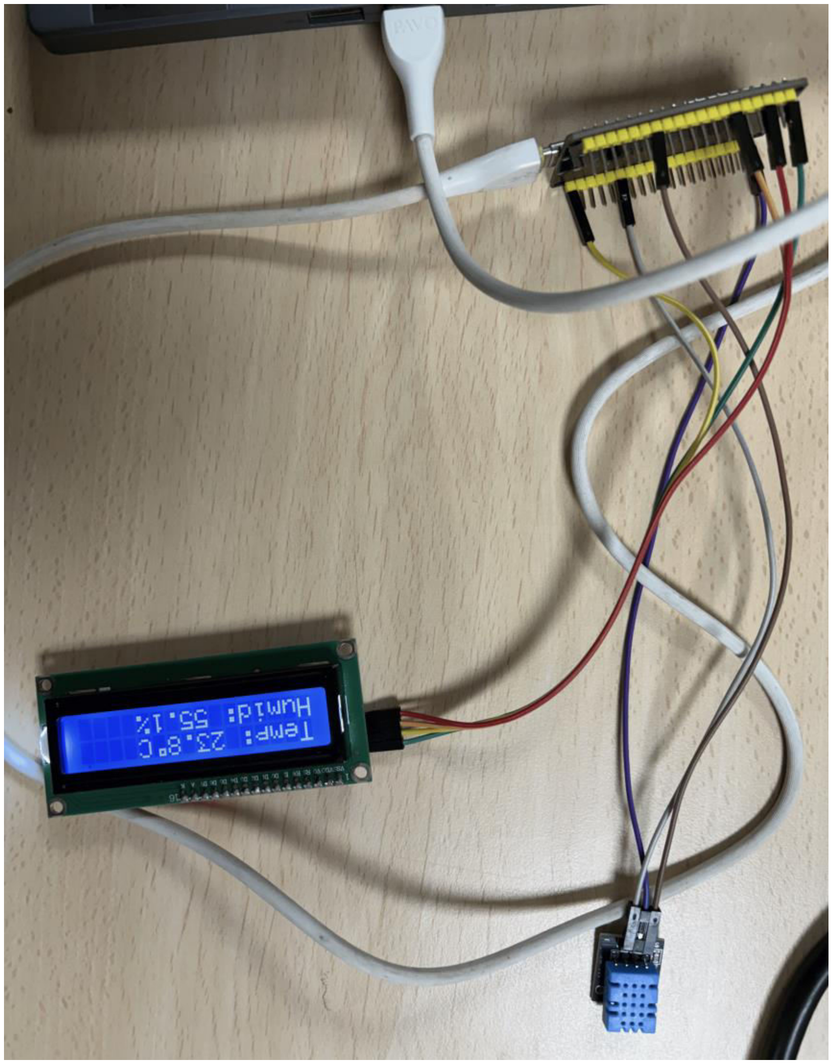

Figure 30: Test 2.1

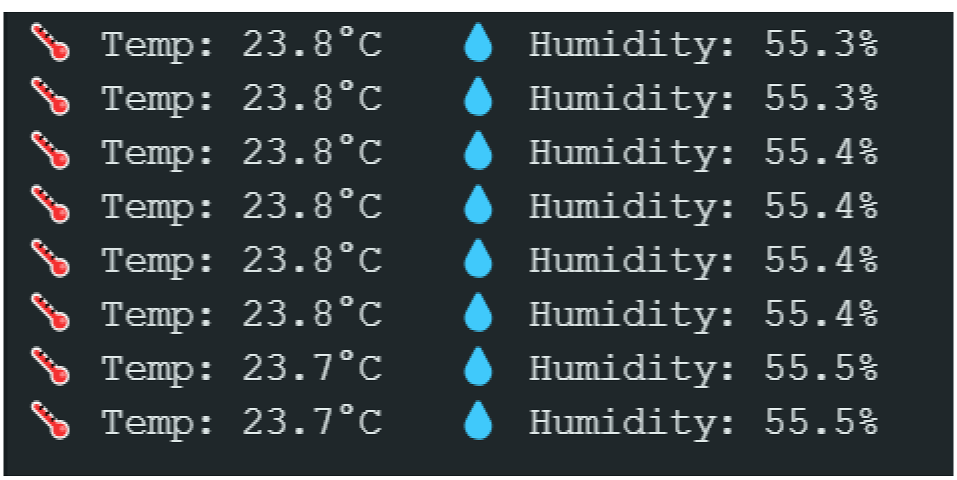

Figure 31: Test 2.2

5.3. Test 3: To Check If the Ultrasonic Distance Sensor Works with Buzzer or Not.

Test	**3**
Objective	To check if the ultrasonic distance sensor works with the buzzer or not.
Activity	The digital pins are connected to the digital pins of the ultrasonic sensor that were connected to digital GPIO pins of ESP. The positive terminal of the buzzer is connected to the digital GPIO pin of ESP.
Expected result	When the distance reading of the ultrasonic sensor is less than or equal to 10cm the buzzer should buzz. The reading should be logged continuously in the terminal.
Actual result	When the distance reading of the ultrasonic sensor was less than or equal to 10cm the buzzer buzzed. The reading was logged continuously in the terminal.
Conclusion	The test was successful

Table 4: Test 3

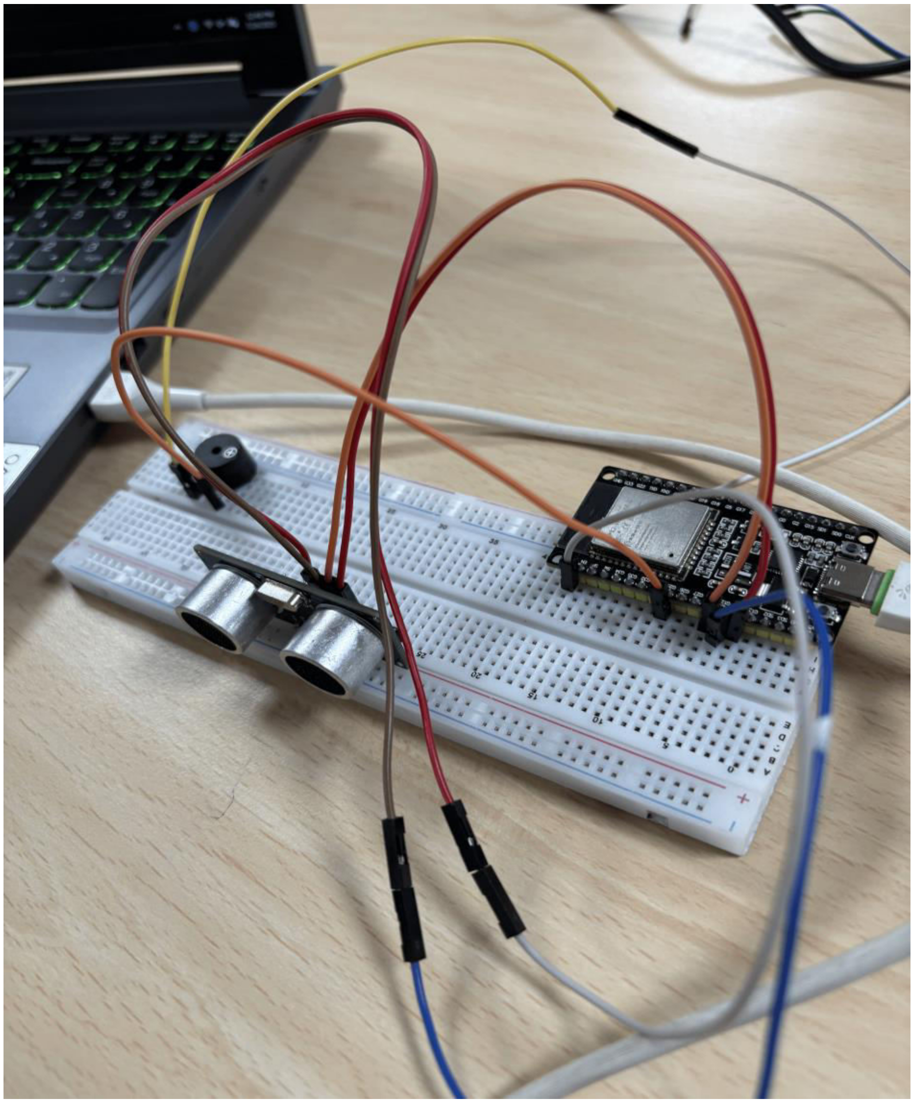

Figure 32: Test 3.1

```
📏 Distance: 75.1 cm ✅ Safe Distance 🔕 Buzzer OFF
📏 Distance: 91.9 cm ✅ Safe Distance 🔕 Buzzer OFF
📏 Distance: 57.3 cm ✅ Safe Distance 🔕 Buzzer OFF
📏 Distance: 49.0 cm ✅ Safe Distance 🔕 Buzzer OFF
📏 Distance: 8.2 cm ⚠️ Too Close! 🔊 Buzzer ON
📏 Distance: 32.1 cm ✅ Safe Distance 🔕 Buzzer OFF
```

Figure 33: Test 3.2

5.4. Test 4: To Check If the Double Relay Works with the Water Pump Motor.

Test	**4**
Objective	To check if the double relay works with the water pump motor.
Activity	The logic voltage for the relay was given by the ESP, the IN1 of the relay was connected to the GPIO pin of ESP and GND was also connected to the ESP's GND rail. The positive terminal of the water pump motor was connected to the normal open of the relay and the common of the water pump motor was connected to the 5V adapters common ground and the relay's common was connected to the positive terminal of the 5V adapter.
Expected result	When the IN1 received Voltage signal the water pump would be normally closed and the water pump would start.
Actual result	When the IN1 received Voltage signal the water pump was normally closed and the water pump started.
Conclusion	The test was successful.

Table 5: Test 4

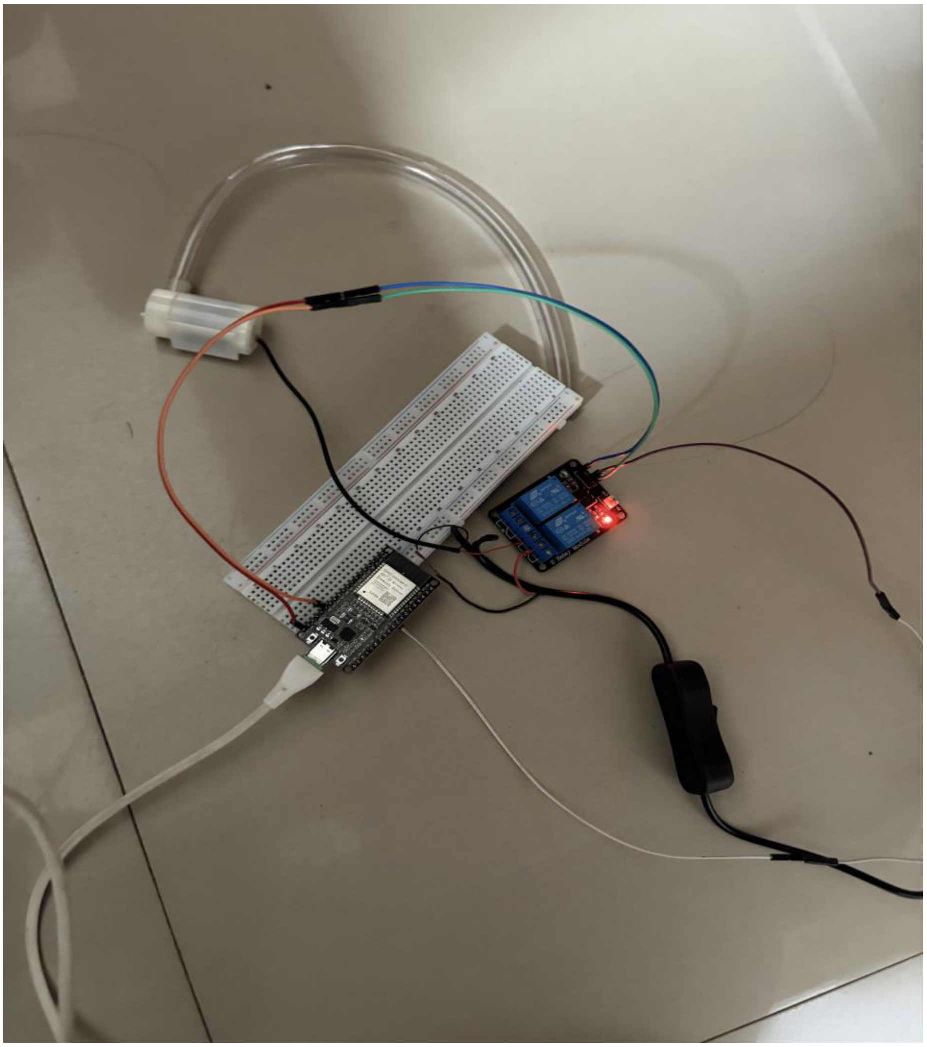

Figure 34: Test 4.1

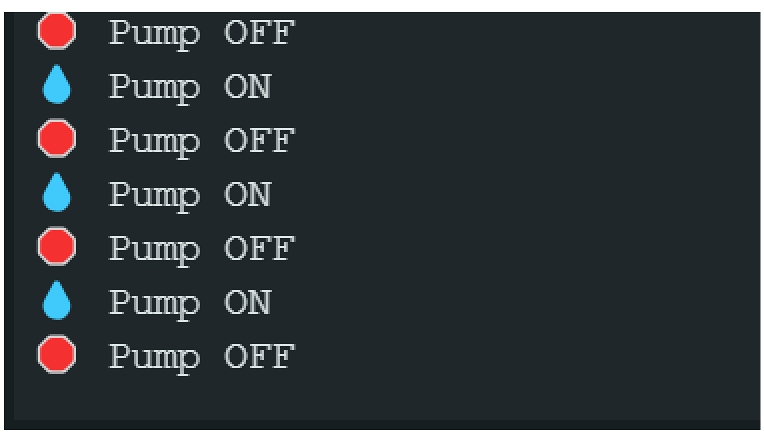

Figure 35: Test 4.2

5.5. Test 5: To check if Blynk is integrated properly with ESP32.

Test	**5**
Objective	To check if the data collected by the sensors using ESP32 displays in the Blynk dashboard.
Activity	Blynk's library and boilerplate were integrated in the original code.
Expected result	The data collected using the sensors should be displayed in the Blynk's dashboard. Also, the water pump should turn on and off when the widget is pressed accordingly.
Actual result	The data collected using the sensors was displayed in Blynk's dashboard. Also, the water pump turned on and off when the widget is pressed accordingly
Conclusion	The test was successful.

Table 6: Test 5

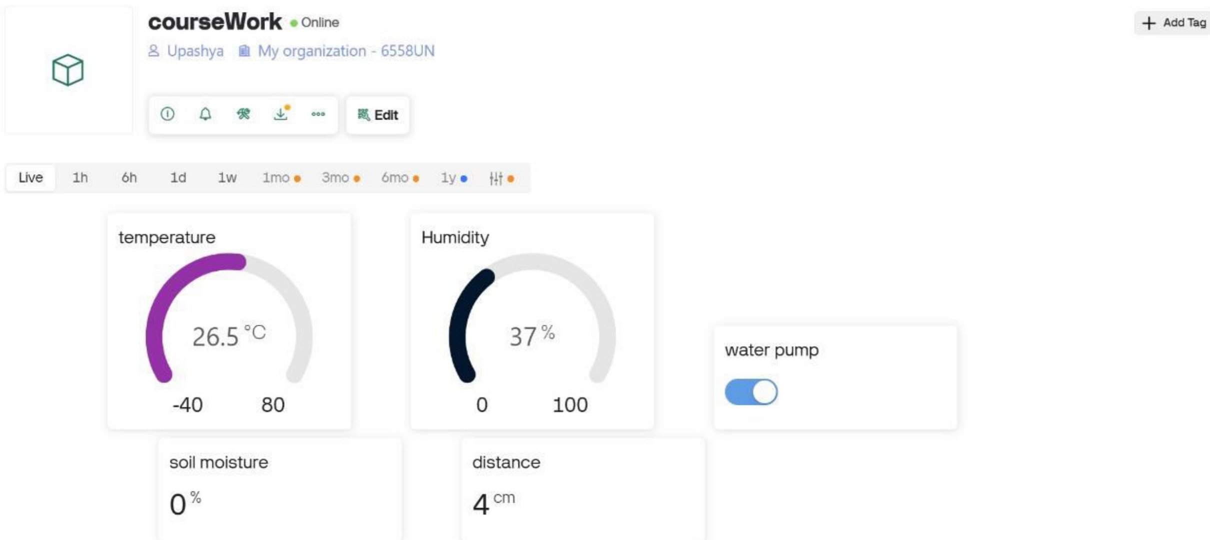

Figure 36: Test 5.1

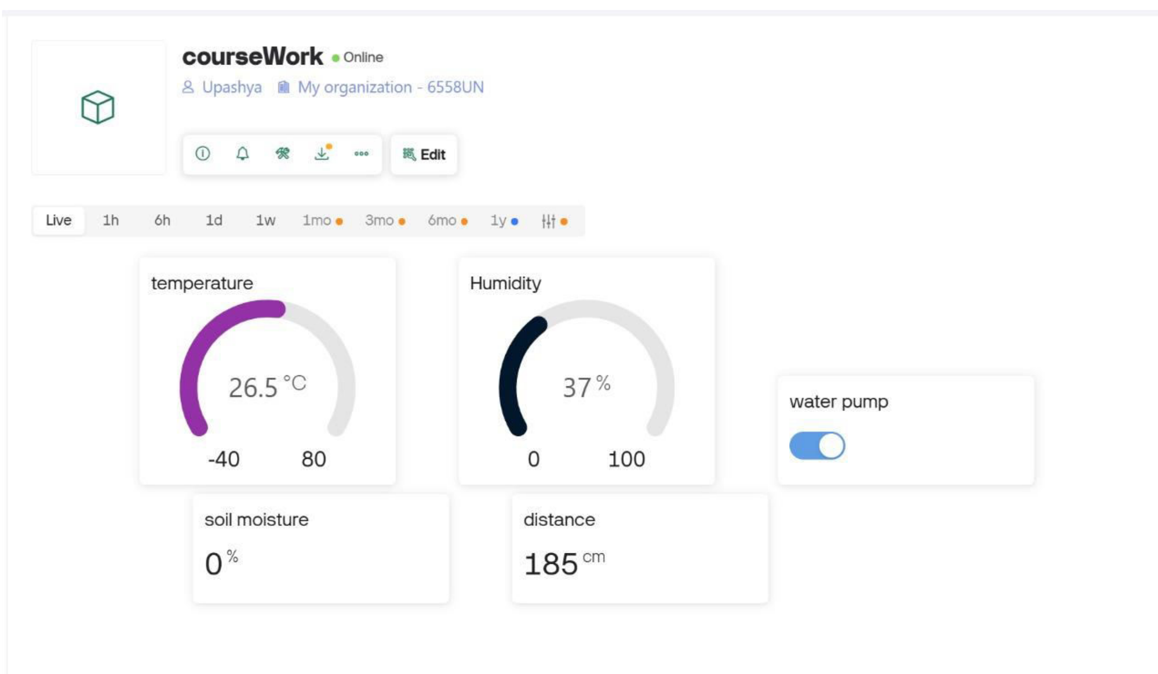

Figure 37: Test 5.2

6. Future Works

Our project is based on implementing IOT systems, which is a rapidly growing technology and is implemented in most sectors of industries across the world (Yadav, 2023). As time passes by a greater number of IOT based systems will be seen, leading to almost all future work being dependent upon such integrated technology (Mushtaq, 2023). The project we have developed so far is just an automatic crop watering concept that could have several additional IOT systems integrated to further help modernize agriculture,

- Develop and join a well to the water tank for continuous water supply that removes the water shortage problem in case tanks run out of water (Lamsal, 2023).
- Implementing intrusion detectors and linking them with a buzzer that sounds when it detects any anomaly gives protection to the crops (Yadav, 2023).
- Soil NPK sensors could be planted to check the fertility of soil through blynk apps and fertilizers could be used accordingly. Similarly, bio sensors and chemical sensors could be used for checking the number of insecticides and pesticides (Akan, 2024).
- Mitigating impacts of weather. For example, using row covers to protect against frost and winds in cold weather (FAO, 2024).
- Restructuring housing for covering IOT components properly to protect them against external environment. Also, creating a larger container for placing soil and a variety of crops (Mushtaq, 2023).

7. Conclusion

While developing our project, we have ensured our system complies with the necessary ethical and legal standards that are demanded to be met while making such projects. The IOT system has been created while keeping in mind the possible harm that could be caused to the external environment (FAO, 2024). It is also essential that the data collected by the IOT system is safe and secure. We confirm users' privacy and make sure no data breaches occur (Yadav, 2023). Data is collected, used, and stored safely following all data protection laws (Giri, 2023). Also, software that has been used such as Blynk platform or Arduino IDE should be verified if they are genuine and properly licensed software's along with whether their terms and conditions have been followed thoroughly (Yadav, 2023). All in all, the main aim of this project is to modernize agriculture using the IOT integrated system. The project uses different sensors, actuators, and microprocessors to automate agriculture based on the readings from the sensors and instructions given to the microprocessor. This helps the farm to be efficient and productive based on when to water plants and how much water without human effort (Babar, 2024) (Akan, 2024). The system created is just a first building step which could lead to several IOT based technologies useful in farming. Overall, our system shows how IOT based technology could be the next big thing in revolutionizing the agricultural industry (Mushtaq, 2023).

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9. Appendix

9.1. Logbook 1

LOGBOOK 1:

ITEMS WE HAVE DISCUSSED:
We discussed what we should be doing and started dividing about which parts of the report we will be taking on. And created a rough sketch for the project.

PROBLEMS WE FACED:
We faced a problem which was communication between the group members. And also to decipher the various previous case studies.

PROBLEMS WE MITIGATED:
We ultimately ended those previous problems by establishing proper communication through platforms like teams and instagram. And ultimately planned to decipher various studies part by part.

ACHIEVEMENT OF THE WEEK:
We managed to complete the first milestone report.

Figure 38: Logbook 1

9.2. Logbook 2

LOGBOOK 2:

ITEMS WE HAVE DISCUSSED:
We requested material through our lecturer and started analysing what function each of the IoT devices perform and discussed it.

PROBLEMS WE FACED:
We faced a problem which was while testing the functionality of the devices one by one the devices started failing.

PROBLEMS WE MITIGATED:
We ultimately ended up switching those faulty devices and acquired better devices to comprehend each task.

ACHIEVEMENT OF THE WEEK:
We managed to distinguish all the devices and their functionality and discovered their applications.

Figure 39: Logbook 2

9.3. Logbook 3

LOGBOOK 3:

ITEMS WE HAVE DISCUSSED:
We started connecting the devices to make a prototype of the project and every member was made to design a circuit to understand.

PROBLEMS WE FACED:
Firstly, the ESP was a faulty one along with the LCD. We tried to work it out for several hours. Secondly, the wires in the water pump were faulty and jumper wires were fully female.

PROBLEMS WE MITIGATED:
We managed to change the ESP, LCD, water PUMP and jumper wires and completed the prototype of the project.

ACHIEVEMENT OF THE WEEK:
We managed to make the prototype and made it work.

Figure 40: Logbook 3

9.4. Logbook 4

LOGBOOK 4:

ITEMS WE HAVE DISCUSSED:
We discussed dividing the report work for the second milestone and also discussed the system for the second milestone along with the codes.

PROBLEMS WE FACED:
There were no peculiar problems this week. But a sense of low confidence did stir up between the members.

PROBLEMS WE MITIGATED:
We talked among ourselves to build up the confidence within the group.

ACHIEVEMENT OF THE WEEK:
We managed to finish the second milestone report with proper evidence and code.

Figure 41: Logbook 4

9.5. Logbook 5

LOGBOOK 5:

ITEMS WE HAVE DISCUSSED:
We started creating the physical model for the system to be placed. It included making a house, a farm and a water source along with some decorations and designs.

PROBLEMS WE FACED:
We faced problems like disagreement in the designs and collected various designs and references ultimately confusing us.

PROBLEMS WE MITIGATED:
We ended up finalizing on a design and started inputting our ideas regarding the project.

ACHIEVEMENT OF THE WEEK:
We created the skeletal version for the house and the farm.

Figure 42: Logbook 5

9.6. Logbook 6

LOGBOOK 6:

ITEMS WE HAVE DISCUSSED:
We contemplated ways to integrate the circuit to the physical model and also connect it to blynk. Along with that we discussed completion of the report with proper formatting and instructions.

PROBLEMS WE FACED:
Some of the description and information on the report were jumbled along with blynk not receiving the data.

PROBLEMS WE MITIGATED:
We managed to complete the report with proper information and description and also managed to solve the issue with blynk.

ACHIEVEMENT OF THE WEEK:
We managed to complete the report work completely for final submission.

Figure 43: Logbook 6

9.7. Source Code

```
#define BLYNK_TEMPLATE_ID "TMPL6Jfw6s8tV"
#define BLYNK_TEMPLATE_NAME "courseWork"
#define BLYNK_AUTH_TOKEN  "kBsd-1nJQkse-1hCp5nugKlSdgnRwmNI"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>
#include <ESP32Servo.h>

// --- WiFi ---
char ssid[] = "Islington College";
char pass[] = "I$LiNGT0N2025";

// --- LCD ---
LiquidCrystal_I2C lcd(0x27, 16, 2);

// --- DHT Sensor ---
#define DHTPIN 4
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);

// --- Soil Moisture ---
#define SOIL_PIN 34
```

```
int soilThreshold = 30;

// --- Relay ---
#define RELAY_PIN 27
bool pumpAutoControl = true;  // Auto ON/OFF

// --- Buzzer ---
#define BUZZER_PIN 14
float tempHighThreshold = 35.0;

// --- Ultrasonic ---
#define TRIG_PIN 26
#define ECHO_PIN 25

int distanceThreshold = 10;

// --- Servo ---
Servo myServo;
#define SERVO_PIN 13

// --- Blynk Virtual Pins ---
#define VPIN_TEMP V0
#define VPIN_HUM  V1
#define VPIN_SOIL V2
#define VPIN_DIST V3
#define VPIN_PUMP V4
```

```
// Timing variables
unsigned long lastSensorRead = 0;
const unsigned long sensorInterval = 2000; // Read sensors every 2 seconds

// --- BLYNK: Pump Control ---
BLYNK_WRITE(VPIN_PUMP) {
  int value = param.asInt();
  pumpAutoControl = (value == 0); // If button is OFF, enable auto control

  if (value == 1) {
    digitalWrite(RELAY_PIN, LOW);  // Active LOW
    Serial.println("🚰 Pump ON (Manual from Blynk)");
  } else {
    digitalWrite(RELAY_PIN, HIGH);
    Serial.println("🚰 Pump OFF - Auto mode enabled");
  }
}

void setup() {
  Serial.begin(115200);

  // Initialize pins first
  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, HIGH); // Pump OFF initially
  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW); // Buzzer OFF initially
```

```
  // Ultrasonic Init
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

  // Servo Init
  myServo.attach(SERVO_PIN);
  myServo.write(0);

  // LCD Init with error checking
  lcd.init();
  lcd.backlight();
  lcd.clear();

  // Sensors Init
  dht.begin();

  // Wait for DHT to stabilize
  delay(2000);

  // Start Blynk (no LCD messages)
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

  Serial.println("🌱 Smart Garden System Started");
  lcd.clear();
}

void loop() {
```

```
Blynk.run();

unsigned long currentMillis = millis();

// Read sensors at intervals to avoid blocking
if (currentMillis - lastSensorRead >= sensorInterval) {
 lastSensorRead = currentMillis;

 // --- Read DHT ---
 float temp = dht.readTemperature();
 float hum = dht.readHumidity();

 // Check if DHT reading is valid
 if (isnan(temp) || isnan(hum)) {
  Serial.println("❌ DHT22 reading failed!");
  temp = 0.0;
  hum = 0.0;
 }

 // --- Read Soil ---
 int soilValue = analogRead(SOIL_PIN);
 int soilPercent = map(soilValue, 0, 4095, 100, 0);
 // Constrain soil percentage
 soilPercent = constrain(soilPercent, 0, 100);

 // --- Read Distance ---
 long duration, distance;
```

```
digitalWrite(TRIG_PIN, LOW);
delayMicroseconds(2);
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);
duration = pulseIn(ECHO_PIN, HIGH, 30000); // 30ms timeout

if (duration == 0) {
  distance = 999; // No echo received
} else {
  distance = duration * 0.034 / 2;
}

// --- Display on LCD (Only Temp & Humidity) ---
lcd.clear(); // Clear display each time
lcd.setCursor(0, 0);
lcd.print("T:");
if (temp > 0) {
  lcd.print(temp, 1);
} else {
  lcd.print("--");
}
lcd.print((char)223);
lcd.print("C H:");
if (hum > 0) {
  lcd.print(hum, 0);
} else {
```

```
    lcd.print("--");
  }
  lcd.print("%");

  // --- Debug Serial ---
  Serial.println("=== Sensor Readings ===");
  Serial.print("🌡 Temp: "); Serial.print(temp); Serial.println(" °C");
  Serial.print("💧 Hum: "); Serial.print(hum); Serial.println(" %");
  Serial.print("🌱 Soil: "); Serial.print(soilPercent); Serial.println(" %");
  Serial.print("📏 Dist: "); Serial.print(distance); Serial.println(" cm");

  // --- Blynk Send (only if connected) ---
  if (Blynk.connected()) {
    Blynk.virtualWrite(VPIN_TEMP, temp);
    Blynk.virtualWrite(VPIN_HUM, hum);
    Blynk.virtualWrite(VPIN_SOIL, soilPercent);
    Blynk.virtualWrite(VPIN_DIST, distance);
  }

  // --- Soil Moisture Pump Auto Control ---
  if (pumpAutoControl) {
    if (soilPercent < soilThreshold) {
      digitalWrite(RELAY_PIN, LOW);
      Serial.println("🚰 Pump ON (Auto - Soil Dry)");
      if (Blynk.connected()) {
        Blynk.virtualWrite(VPIN_PUMP, 1);
```

```
      }
    } else {
      digitalWrite(RELAY_PIN, HIGH);
      Serial.println("🚰 Pump OFF (Auto - Soil OK)");
      if (Blynk.connected()) {
        Blynk.virtualWrite(VPIN_PUMP, 0);
      }
    }
  }

  // --- Temp High Buzzer ---
  if (temp > tempHighThreshold && temp > 0) {
    digitalWrite(BUZZER_PIN, HIGH);
    Serial.println("🔔 Temp HIGH Alarm!");
  } else {
    digitalWrite(BUZZER_PIN, LOW);
  }

  // --- Ultrasonic Servo ---
  if (distance > 0 && distance < distanceThreshold && distance != 999) {
    myServo.write(90);
    Serial.println("🔧 Servo Open");
  } else {
    myServo.write(0);
    Serial.println("🔧 Servo Closed");
  }
}
```

```
  delay(100); // Small delay to prevent watchdog reset
}
```